Effect of Surface Defects and Thermochemistry on Laminar Hypersonic Boundary Layers

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The present research work proposes a series of computational fluid dynamics (CFD) simulations performed in STAR-CCM+ to investigate the effects of various surface defect depths and thermochemistry models on hypersonic boundary layer development and surface heating effects. Surface defects, large or small, can significantly impact the flow thermodynamics under hypersonic conditions. Accurate simulation results and high-fidelity CFD frameworks are crucial to predicting these effects on boundary layer structure, surface heating, and ablation for hypersonic thermal protection systems. The defects analyzed in this investigation were hyperbolic tangent divots of heights $h = 4$, $h = 8$, and $h = 12$. Steady, laminar, ideal gas model simulations were run on all three divot heights. Additionally, two real gas models were run on the $h = 12$ divot geometry to assess their impact on boundary layer structure and heating. The results indicate a peak temperature drop in the $h = 8$ divot attributed to energy dissipation from a dominant recirculating vortex, while the $h = 4$ and $h = 12$ divots exhibit comparable maximum temperatures. The $h = 12$ divot additionally exhibits two recirculation zones, suggesting a transition to a closed cavity flow regime. Gas model comparisons reveal that the ideal and non-equilibrium air models produce nearly identical surface heat transfer distributions, while the equilibrium air model predicts consistently elevated heating across the divot surface. These findings highlight the sensitivity of hypersonic surface heating predictions to both defect geometry and gas model selection, motivating further investigation into the governing thermochemical and flow structural mechanisms.

I. Introduction

Modeling and simulation of hypersonic flows has been and continues to be a significant challenge and active area of research within the world of computational fluid dynamics. Many of the difficulties arise when trying to predict nonequilibrium aerothermodynamic flow properties such as high-energy molecular interactions, gas ionization, electron excitation, strong shock formation, and species dissociation. These phenomena are highly volatile and can significantly impact surface ablation and heating rates on a hypersonic vehicle. Obtaining an accurate prediction of hypersonic flow-to-surface interactions is a critical part of developing vehicles that must operate in this regime. In addition to this, surface defects on hypersonic flow surfaces can substantially impact aerodynamic performance. Surface defects can arise from various sources, including atmospheric debris collisions, manufacturing tolerance errors, and surface ablation during reentry. Surface irregularities of this nature on hypersonic surfaces can trigger local flow separation, unwanted recirculation and shock-boundary layer interactions, all of which can dramatically impact surface heating. These effects can still be substantial even if the surface is only slightly perturbed. Among the most common surface defect geometries are divots: shallow cavities that can form on the surface of a hypersonic vehicle and produce highly localized flow disturbances. Despite their relatively simple geometry, divots can generate complex recirculation structures and significant local heating augmentation, yet their effects remain understudied relative to other surface defect types. Computational fluid dynamics provides a practical and cost-effective means of investigating these phenomena, offering flexibility in geometry and flow conditions that would be prohibitively expensive or time-consuming to replicate experimentally. In this work, CFD simulations are performed to assess the effects of varying divot depth on boundary layer development and surface heating, as well as the sensitivity of these predictions to the choice of gas and chemistry model.

Previous computational and experimental studies have examined the effects of surface roughness and geometric irregularities on hypersonic boundary layer transition and heating augmentation [1]. The effects of surface roughness

elements on hypersonic boundary layer development and heating have also been investigated, with studies of rectangular and cylindrical geometries across varying Mach numbers demonstrating significant sensitivity of the boundary layer to surface irregularities [2]. More recently, the divot geometry considered in the present work was employed by LeFebvre, Orfaly, and Massa to investigate anomalous ablation rates in hypersonic boundary layers, highlighting the complex thermal interactions that arise in this configuration [3]. Building on this, Lavisetty and Massa examined viscous-inviscid interactions in hypersonic boundary layers over surface defects, demonstrating the significant influence of defect geometry on local flow structure and surface heating [4]. Despite these contributions, the effects of systematically varying divot depth and gas model fidelity on boundary layer development and surface heating remain limited, motivating the present investigation.

The abstract is organized as follows. Section II discusses the software and simulation set-up used for this investigation. Section III presents the results of three different divot height simulations, as well as the effects of three different gas models on a single divot geometry. Section IV discusses the preliminary conclusions and future objectives of the experiment.

II. Computational Methodology

Siemen's STAR-CCM+ was used for both the mesh generation and the flow simulations [5]. STAR-CCM+ is an interactive, GUI-controlled computational fluid dynamics solver with numerous, highly customizable physics and chemistry models to accurately predict a variety of flow scenarios. STAR-CMM+ has previously been validated as a reputable method for predicting hypersonic flows in prior publications. Cross and West [6] validated hypersonic flow results in STAR-CCM+ over various geometries through comparison to real wind tunnel experiments. They found that while the simulated predictions were not in exact agreement with experimental data, they proved sufficient to validate the hypersonic capabilities of the software.

A. Domain

The chosen geometry in this investigation was a flat plate with a small, perturbed defect located in the center. In order to maintain a smooth curve and avoid discontinuities, a hyperbolic tangent equation obtained from Lavisetty and Massa [4] was used to generate the divots. The hyperbolic tangent equation used for the divot profiles is as follows:

$$y = \frac{h}{2\tanh(d)}(\tanh(mx + d) - \tanh(mx - d)) \tag{1}$$

Where $d = 0.5$, $m = 5$, and h is the height of the divot that is varied throughout the investigation. The output y values are scaled by $(10^8)^{-\frac{1}{4}}$ to smooth out the divot profile and prevent high, narrow spiking towards the center. The divot profiles were drawn over a dimensionless x-range of [-3, 3] in order to provide a sufficient flat-plate surface area before and after the defect formation. Walls with a dimensionless height of 2 were drawn on each end of the plate which were then connected by the top wall to close out the geometry. This formed a large, rectangular domain with the bottom surface face slightly perturbed by the hyperbolic tangent geometry. These shapes were generated in Python and exported to STAR-CCM+ as .STL files where inlet, outlet, surface wall, and symmetry plane regions were defined.



Fig. 1 Computational Domain Geometry

B. Physics Models and Boundary Conditions

Specific physics models, boundary conditions, and initial conditions were carefully chosen to best capture the desired outcomes of the simulation. All preliminary results were generated under steady laminar assumptions for simplicity, time constraints, and computational resource availability. An unsteady simulation was attempted to investigate the potential need for such a solver, but it was found that the flow field behaved in a physically steady manner. Every simulation was run with a coupled energy and flow convection scheme, as opposed to segregated as it is the optimal choice for high-speed, compressible flows with strong physical coupling. For three of the simulations, ideal gas assumptions were enforced to obtain empirical solutions to the flow field. For the h=4 divot height, two more simulations were run under real gas and non-equilibrium gas assumptions to analyze their effects (or lack of) on the flow solution. The boundary conditions were chosen to emulate a Mach 6 hypersonic reentry at an altitude of approximately 20 kilometers. The reference and initial temperature, density, and pressure values were selected using the standard atmosphere table from Anderson's Introduction to Flight [7]. The reference and initial conditions set for each simulation are summarized in the table below:

Parameter	**Value**
Reference Pressure	1 atm
Reference Temperature	300 K
Reference Density	0.08891 kg/m^3
Initial Pressure	5590 Pa
Initial Temperature	216.7 K
Initial Velocity	1770 m/s
Inlet Mach Number	6
Divot Surface Temperature	1000 K

Table 1 Boundary Condition and Reference Parameters

C. Mesh Configuration and Generation

The mesh configuration utilized the same procedure for generation throughout the three divot heights analyzed. The 2-D automated mesh feature built into STAR-CCM+ was used to automatically generate an optimized mesh per given

user input commands. Three bodies were used to generate and refine the polyhedral computational meshes: the divot domain, a volumetric cylinder, and a volumetric sphere. The cylinder and sphere were used to impart a volumetric control on the freestream flow around the divot, and inside the divot itself, respectively. The default computational cell base size was set to 2.6×10^{-4}m, which was roughly the length of the divots. A surface control was enforced on the flat plate and divot wall surface in order to refine the area for better computational results. The cylinder and sphere volumetric controls around the divot area were imparted to further refine the areas of interest in this investigation. The surface and volumetric controls are customized by setting target and minimum cell sizes to user-specified percentages of the established base size. The following table summarizes the used meshing parameters and control values, and the final generated mesh is shown below:

Default Control	**Value**
Base Size	2.6×10^{-4}m
Target Surface Size	10%
Surface Control	**Value**
Target Surface Size	0.5%
Minimum Surface Size	0.1%
Cylindrical Control	**Value**
Custom Size	10%
Spherical Control	**Value**
Custom Size	0.5%
Final Cell Count:	482,188

Table 2 Mesh Control and Refinement Parameters

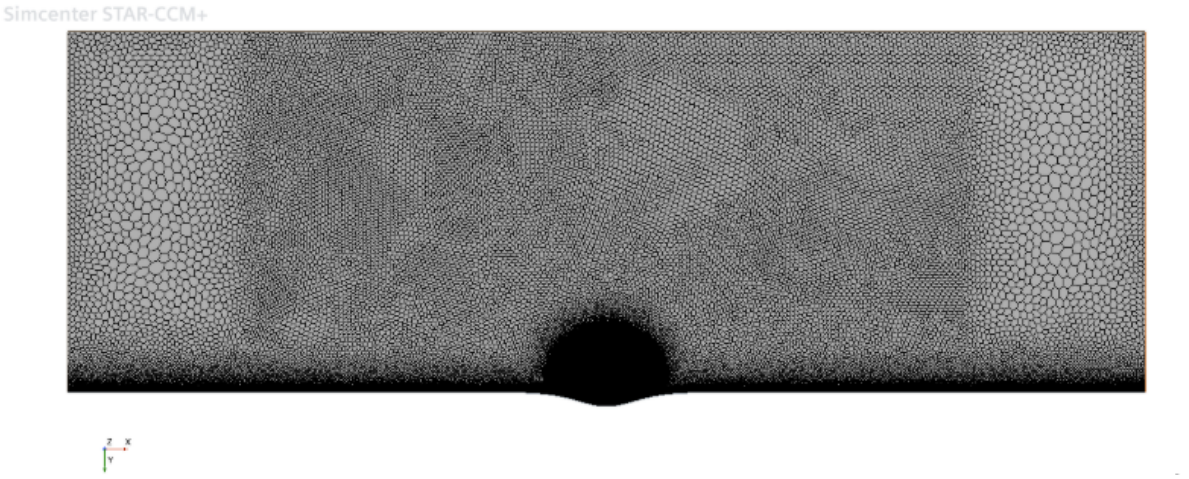


Fig. 2 Polyhedral Computational Mesh

III. Preliminary Results and Discussion

This section presents a discussion on the preliminary simulation results obtained and future objectives of the investigation. Observations and comparisons of the generated flow fields are discussed in detail to provide an in-depth analysis of the results found.

A. Divot Height Comparison - Ideal Gas

Three divot heights were tested in order to assess their effects on the hypersonic flow field and surface boundary layer. The heights of the divots were varied by changing the h parameter in equation 1. The three heights selected were $h = 4$, $h = 8$, and $h = 12$. The results of the simulations are summarized in the following table and figures:

Divot Height (h)	Maximum Temperature (K)	# Recirculation Vortices
4	1311.81	1
8	1287.75	1
12	1318.56	2

Table 3 Maximum temperature and number of recirculation vortices for different divot heights.

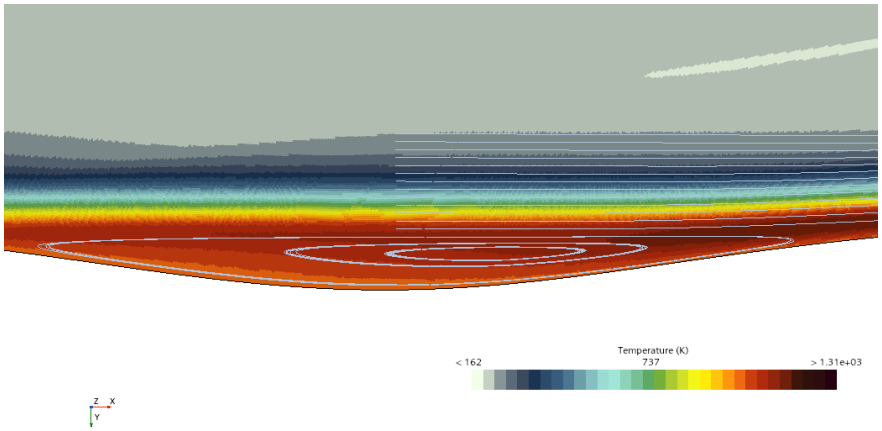


Fig. 3 Temperature contour for divot height $h = 4$.

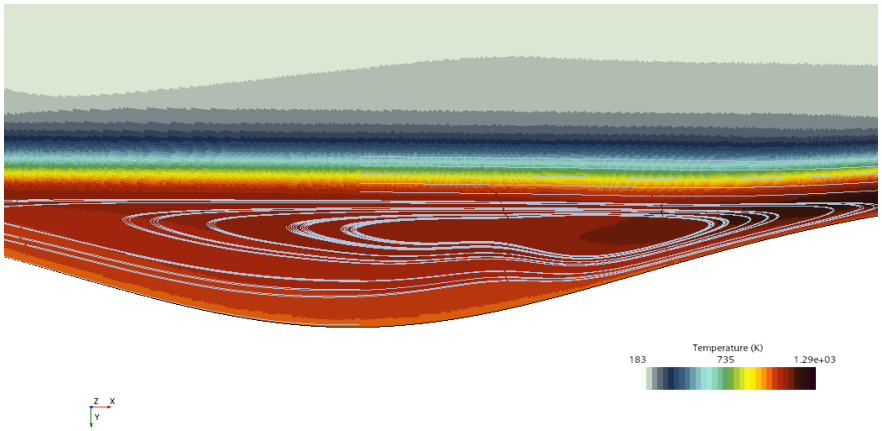


Fig. 4 Temperature contour for divot height $h = 8$.

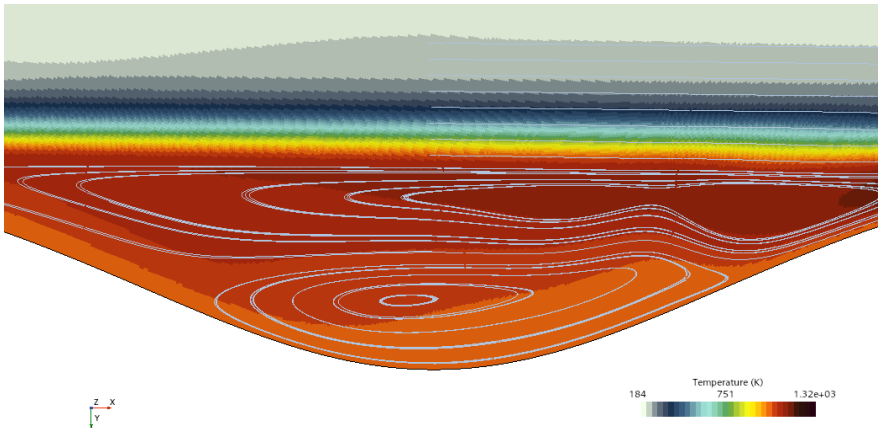


Fig. 5 Temperature contour for divot height $h = 12$.

The $h = 4$ and $h = 12$ divots exhibit comparable maximum temperatures; however, a notable decrease is observed at $h = 8$. This non-monotonic behavior is attributed to differences in recirculation structure within the cavity. At $h = 8$, a single recirculating vortex dominates the divot, where intense shear drives turbulent dissipation and reduces the energy available for surface heating. At $h = 12$, two recirculation zones are observed within the cavity, consistent with a transition to a closed cavity regime. The rotational energy is divided between these two structures, and their interaction likely enhances mixing with the freestream rather than shielding the surface, resulting in a recovery of maximum temperature toward values comparable to those seen at $h = 4$.

B. Gas Model Comparison - h=12 Divot

In addition to assessing the effects of varied divot geometry on boundary layer development and surface heat transfer, several gas and chemistry models were evaluated using the $h = 12$ simulation. Of the three geometries considered, the $h = 12$ divot produced the most severe heating and boundary layer disruption, suggesting that variations in the gas model would produce the most discernible differences in this configuration. As mentioned previously, the baseline simulations for all three divot heights used an ideal gas model. While this model provides a reasonable, computationally efficient estimate of the flow field, it neglects several real-gas effects that become increasingly significant at hypersonic speeds, including vibrational and electronic excitation, species dissociation and recombination, and thermochemical non-equilibrium. To assess the influence of these effects on the predicted heating and flow structure, two additional models were applied to the $h = 12$ geometry: a real gas model and a non-equilibrium ideal air model. The results of the three simulations are summarized and discussed below:

Gas Model	Maximum Temperature (K)	# Recirculation Vortices
Ideal	1318.56	2
Equilibrium Air (Real)	1203.24	1
Non-Equilibrium Ideal	1310.90	2

Table 4 Maximum temperature and number of recirculation vortices for different gas models.

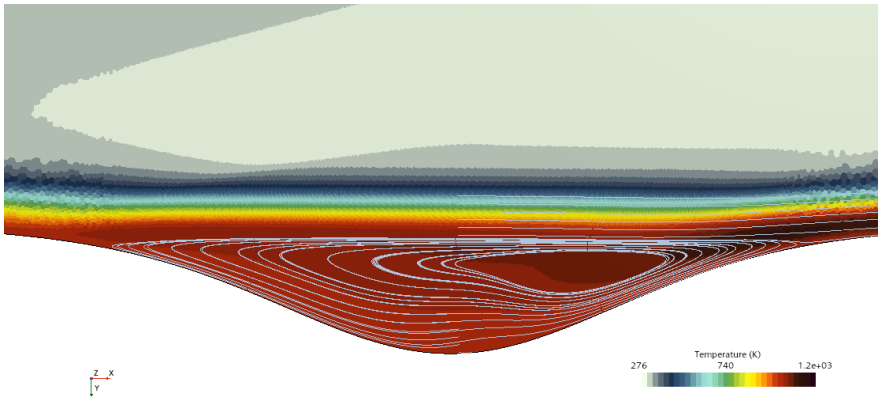


Fig. 6 Real gas temperature contour for h = 12 divot

In the equilibrium real gas model simulation, only one recirculation region was observed as opposed to the previously observed two (see fig.6). This suggests that the modified thermodynamic properties of the real gas model alter the recirculation structure within the cavity by diverting energy to real-gas processes. Additionally, a lower maximum temperature was observed in the real gas simulation, with a discrepancy of approximately 120 K. To further analyze the various effects of the different gas models on the flow thermodynamics, the localized boundary heat transfers from each simulation were plotted against each other to point out further possible discrepancies:

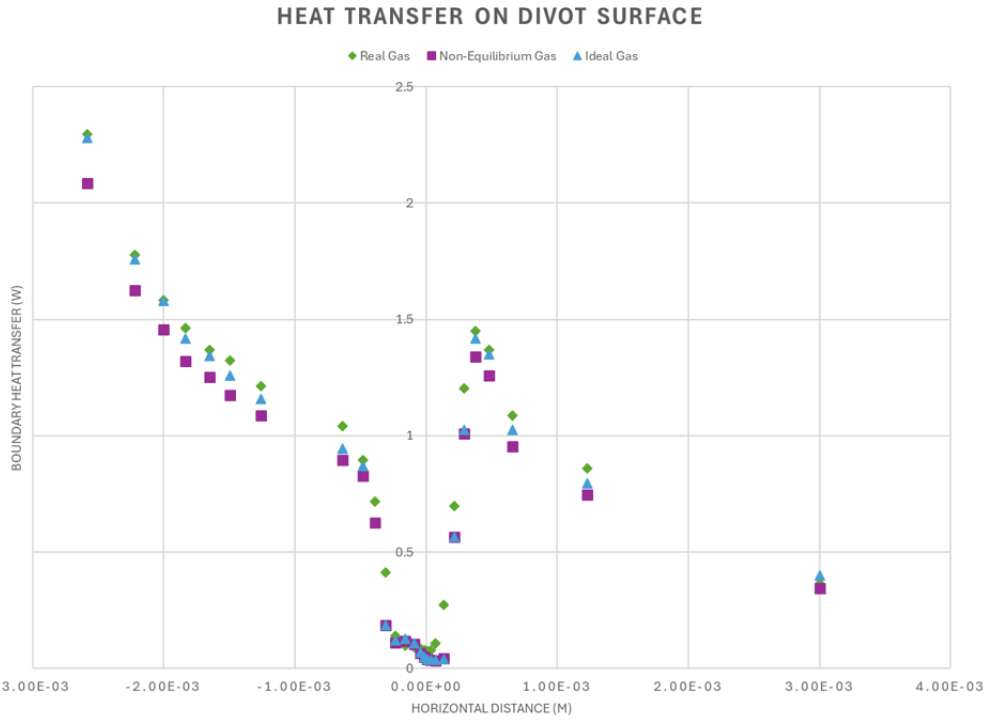


Fig. 7 Local boundary heat transfer on the divot surfaces simulated using three different gas models

All three models produce qualitatively consistent distributions, with peak heat transfer occurring at the upstream lip of the divot and a minimum near the divot floor, consistent with the presence of a recirculating flow structure that shields the surface from the freestream. The ideal and non-equilibrium air models produce nearly identical heat transfer distributions across the entire surface, suggesting that the flow resides in a chemically frozen regime in which finite reaction rates prevent significant thermochemical activity over the relevant flow timescales. The equilibrium air model,

however, predicts consistently higher heat transfer across the divot surface, with the largest discrepancies observed at the upstream lip and the downstream reattachment region. These differences are attributed to the modified thermodynamic properties assumed by the equilibrium model, though the precise source of this behavior warrants further investigation.

IV. Conclusions and Future Work

Preliminary CFD simulations were performed in STAR-CCM+ to assess the effects of divot depth and gas model selection on boundary layer development and surface heating effects in a Mach 6 hypersonic flow. Current results indicate that the depth of the divot significantly influences the local flow recirculation structure as well as the peak surface temperature. The $h = 8$ divot produced a notable reduction in maximum temperature attributed to enhanced turbulent dissipation from a single recirculation vortex. Comparisons of the gas models on the $h = 12$ divot revealed that the ideal and non-equilibrium air models produce nearly identical surface heat transfer distributions, while the real-gas equilibrium air model predicted elevated surface heat transfer. Additionally, the peak temperature decreased by roughly 120 K and only one recirculation region was observed. These observations are likely attributed to energy dissipation due to real-gas effects. The ideal gas and non-equilibrium models produced near identical results, suggesting either a chemically frozen region within the divot or low fidelity of the non-equilibrium solver in STAR-CCM+, warranting further investigation. Future work will include additional flow visualization and thermochemical analysis to better characterize the recirculation structures and validate gas model selection methodology, as well as expanded simulations across a broader range of divot geometries and flow conditions with possible parameterization.

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